INTEGRATING PROBLEMS

10–52 A Thévenin's Theorem from Time-Domain Data
A black box containing a linear circuit has an on/off switch and a pair of external terminals. When the switch is turned on, the open-circuit voltage between the external terminals is observed to be

$$v_{OC}(t) = (18e^{-10t} - 12e^{-50t})u(t)\ \text{V}$$

The short circuit current was observed to be

$$i_{SC}(t) = [(138.5e^{-10t} - 174.3e^{-50t} + 32.9e^{-96.6t})]u(t)\ \text{mA}$$

A 50-Ω load resistance is connected across the terminals and the switch turned on again. What is the voltage delivered to the load?

10–53 Design a Load Impedance
In order to match the Thévenin impedance of a source, the load impedance in Figure P10–53 must be

$$Z_L(s) = \frac{s+5}{s+10}$$

(a) What impedance $Z_2(s)$ is required?

(b) How would you realize $Z_2(s)$ using only resistors, inductors, and/or capacitors?(*Hint:* Write $Z_L(s)$ as a sum of admittances, then solve for $Y_2(s)$.)

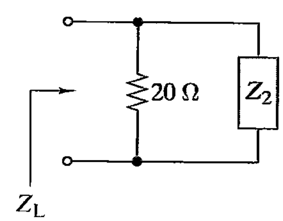


FIGURE P10–53

10–54 *RC* Circuit Analysis and Design
The *RC* circuits in Figure P10–54 represent the situation at the input to an oscilloscope. The parallel combination of R_1 and C_1 represents the probe used to connect the oscilloscope to a test point. The parallel combination of R_2 and C_2 represents the input impedance of the oscilloscope.

(a) Assuming zero initial conditions, transform the circuit into the *s* domain and find the relationship between the test point voltage $V_S(s)$ and the voltage $V_O(s)$ at the oscilloscope's input.

(b) For $R_2 = 10\,\text{M}\Omega$ and $C_2 = 2\,\text{pF}$, determine the values of R_1 and C_1 that make the input voltage a scaled duplicate of the test point voltage.

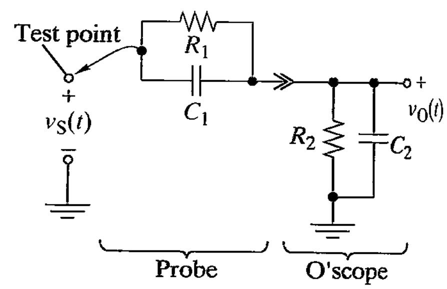


FIGURE P10–54

10–55 *s*-Domain OP AMP Circuit Analysis
The OP AMP circuit in Figure P10–55 is in the zero st… Transform the circuit into the *s* domain and use the OP A… circuit analysis techniques developed in Sect. 4–4 to find t… relationship between the input $V_1(s)$ and the output V_2…

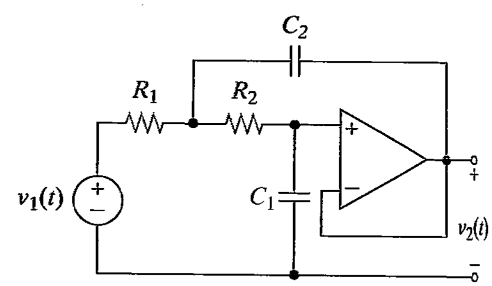


FIGURE P10–55

10–56 Pulse Conversion Circuit
The purpose of the test setup in Figure P10–56 is to deli… damped sine pulses to the test load. The excitation com… from a 1-Hz square wave generator. The pulse conversi… circuit must deliver damped sine waveforms with $\zeta < 0.5$ a… $\omega_0 > 10$ krad/s to 50-Ω and 600-Ω loads. The recommend… values for the pulse conversion circuit are $L = 10\,\text{mH}$ a… $C = 100\,\text{nF}$. Verify that the test setup meets the specific… tions. (*Hint:* Compute the voltage across the load for an inp… signal equal to a unit step function, $u(t)$, and then again for … negative unit step function, $-u(t)$. Note that the output of … square wave generator is the sum of a series of step functions…

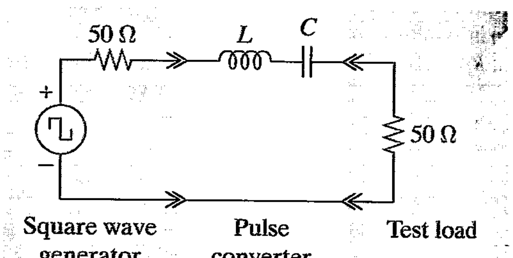


FIGURE P10–56

zero-state response $y(t)$. Symbolically the convolution integral is represented by $y(t) = h(t) * x(t)$. Time-domain convolution and s-domain multiplication are equivalent; that is, $y(t) = h(t) * x(t) = \mathscr{L}^{-1}\{H(s)X(s)\}$. The geometric interpretation of t-domain convolution involves four operations: reflecting, shifting, multiplying, and integrating.

- First- and second-order transfer functions can designed using voltage dividers and inverting or no inverting OP AMP circuits. Higher-order trans functions can be realized using a cascade connecti of first- and second-order circuits. Prototype desi usually require magnitude scaling to obtain practi element values.

PROBLEMS

OBJECTIVE 11–1 NETWORK FUNCTIONS (SECTS. 11–1, 11–2)

(a) Given a linear circuit, find specified network functions and locate their poles and zeros.
(b) Given a linear circuit, select the element values to produce specified poles and zeros. See Examples 11–2, 11–3, 11–4, 11–5, 11–6, 11–7 and Exercises 11–2, 11–3, 11–4, 11–5, 11–6, 11–7

11–1 Find the driving-point impedance seen by the voltage source in Figure P11–1 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

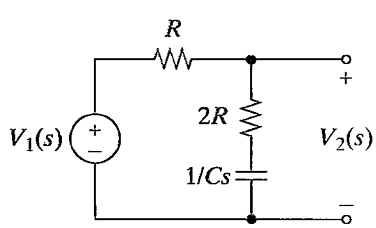


FIGURE P11–1

11–2 Find the driving-point impedance seen by the voltage source in Figure P11–2 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

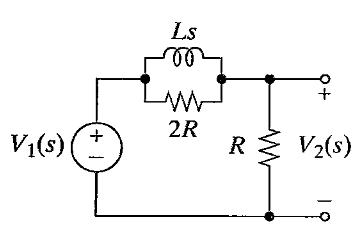


FIGURE P11–2

11–3 Find the driving-point impedance seen by the voltage source in Figure P11–3 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

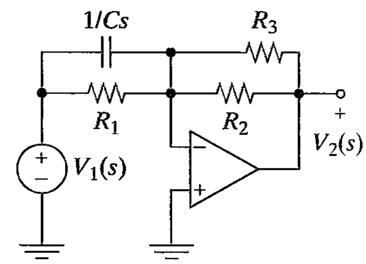


FIGURE P11–3

11–4 Find the driving-point impedance seen by the voltage source in Figure P11–4 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

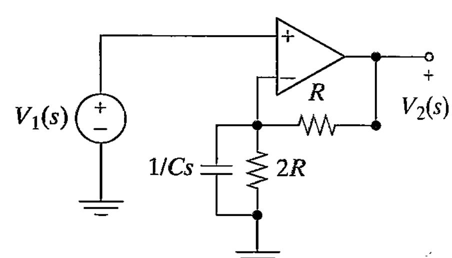


FIGURE P11–4

11–5 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ in Figure P11–5.

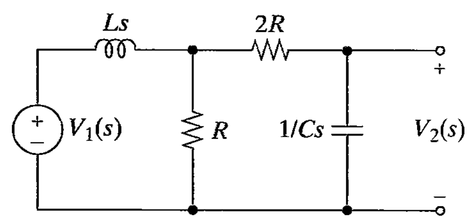


FIGURE P11–5

11-6 Find the driving-point impedance seen by the voltage source in Figure P11–6 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$. Insert a follower at point A and repeat.

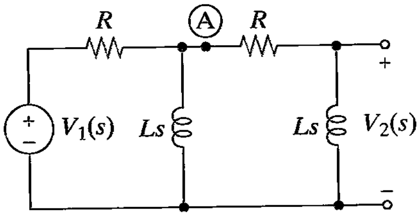


FIGURE P11–6

11-7 Ⓓ Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ for Figure P11–7. Select values of R and C so that $T_V(s)$ has a pole at $s = -1$ krad/s.

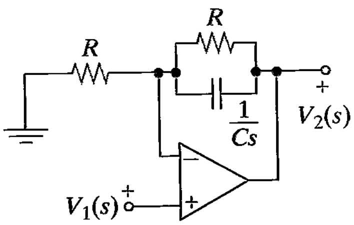


FIGURE P11–7

11-8 Ⓓ Find the current transfer function $T_I(s) = I_2(s)/I_1(s)$ in Figure P11–8. Select values of R and L so that $T_I(s)$ has a pole at $s = -500$ rad/s.

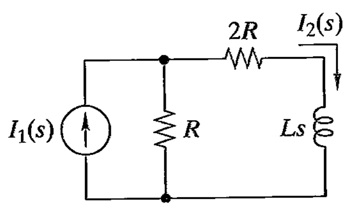


FIGURE P11–8

11-9 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ of the cascade connection in Figure P11–9. Locate the poles and zeros of the transfer function.

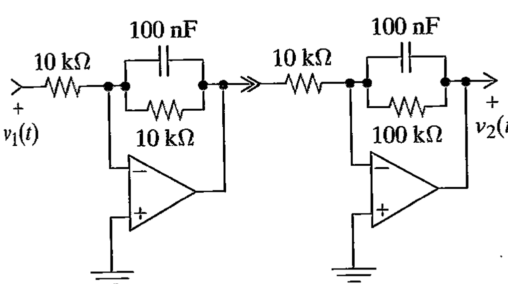


FIGURE P11–9

11–10 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ of the cascade connection in Figure P11–10. Locate the poles and zeros of the transfer function.

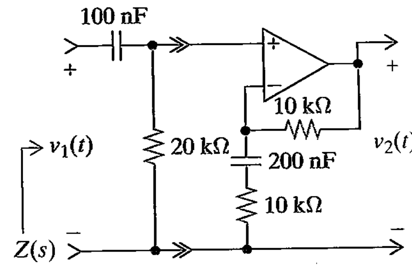


FIGURE P11–10

11–11 Find the input impedance $Z(s)$ in Figure P11–10.

OBJECTIVE 11–2 NETWORK FUNCTIONS, IMPULSE RESPONSE, AND STEP RESPONSE (SECTS. 11–3, 11–4)

(a) Given a first- or second-order linear circuit, find its impulse or step response.
(b) Given the impulse or step response of a linear circuit, find the network functions.
(c) Given the impulse or step response of a linear circuit, find the response due to other inputs.

See Examples 11–8, 11–9, 11–10, 11–11, 11–12 and Exercises 11–8, 11–9, 11–10, 11–11, 11–12, 11–13

11–12 Find $v_2(t)$ in Figure P11–12 when $v_1(t) = \delta(t)$. Repeat for $v_1(t) = u(t)$.

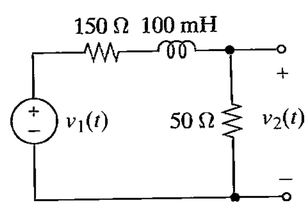


FIGURE P11–12

11–13 Find $v_2(t)$ in Figure P11–13 when $v_1(t) = \delta(t)$. Repeat for $v_1(t) = u(t)$.

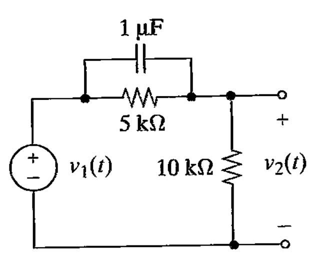


FIGURE P11–13

11–27 The circuit in Figure P11–27 is in the steady state with $v_1(t) = 10\cos 2000t$ V. Find $v_{2SS}(t)$. Repeat for $v_1(t) = 3\cos 10kt$ V.

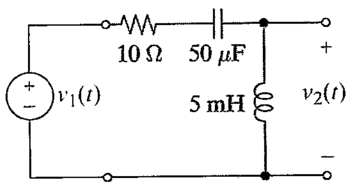


FIGURE P11–27

11–28 The circuit in Figure P11–28 is in the steady state with $i_1(t) = 10\cos 500t$ mA, $R_1 = 100\,\Omega$, $R_2 = 400\,\Omega$, and $L = 100$ mH. Find $i_{2SS}(t)$. Repeat for $i_1(t) = 10\cos 5000t$ mA.

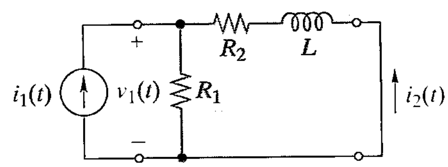


FIGURE P11–28

11–29 The OrCAD circuit drawn in Figure P11–29 is in the steady state with $i_1(t) = 5\cos 1000t$ mA, $R = 1\,\text{k}\Omega$, $L = 2$ H, and $C = 500$ nF. Use OrCAD to find $i_{2SS}(t)$. Copy the figure in OrCAD. Draw your circuit and use IAC for a source. The IPRINT element on the right side of the figure is an ammeter and it comes from the Special Library. It has a polarity as indicated by the small "–" on the element. You should ensure that it will read the correct current direction. Before you can use the IPRINT element, it must be set up to display the current magnitude and phase. Double-click on the element and in its Property Editor place a "y" in the AC, MAG, and PHASE boxes. Click the Apply button and close the Property Editor. To obtain the simulation, select AC Sweep/Noise. Set the start frequency at 159 Hz (1000 rad/s) and the stop frequency at the same value. Place a 1 in Points/Decade. Run the simulation. Under View, select Output File. Scroll down until you see the IPRINT output. Repeat for $i_1(t) = 5\cos 10kt$ mA.

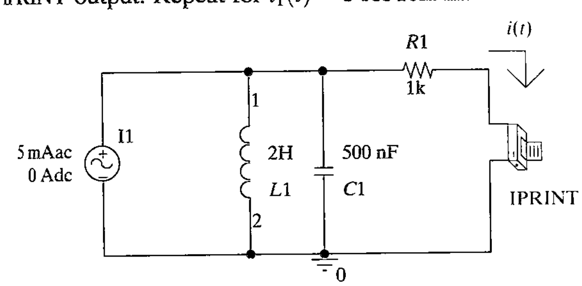


FIGURE P11–29

11–30 The circuit in Figure P11–30 is in the steady state with $i_1(t) = 10\cos 5000t$ mA. Find the steady-state voltage $v_{2SS}(t)$. Repeat for $i_1(t) = 5\cos 2500t$ mA.

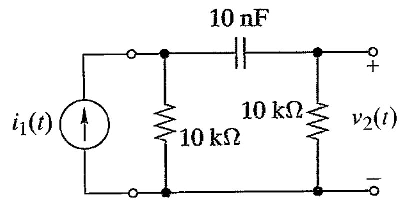


FIGURE P11–30

11–31 The transfer function of a linear circuit is $T(s) = (s + 500)/(s + 50)$. Find the sinusoidal steady-state output for the input $x(t) = 5\cos 500t$.

11–32 The step response of a linear circuit is $g(t) = [5e^{-500t}]u(t)$. Find the sinusoidal steady-state output for the input $x(t) = 2\cos 1000t$.

11–33 The step response of a linear circuit is $g(t) = [2e^{+100t}]u(t)$. Find the sinusoidal steady-state output for the input $x(t) = 5\cos 500t$.

11–34 The impulse response of a linear circuit is $h(t) = [800e^{-1000t}]u(t) - \delta(t)$. Find the sinusoidal steady-state output for the input $x(t) = 3\cos 400t$.

11–35 The impulse response of a linear circuit is $h(t) = 400[e^{-100t} - e^{-400t}]u(t)$. Use MATLAB to find the sinusoidal steady-state output for the input $x(t) = 5\cos 200t$. Use MATLAB to plot $y(t)$.

11–36 The step response of a linear circuit is $g(t) = [e^{-60t}\sin 80t]u(t)$. Find the sinusoidal steady-state response for the input $x(t) = 20\cos 100t$.

11–37 The step response of a linear circuit is $g(t) = [1 - 20te^{-10t}]u(t)$. Find the sinusoidal steady-state response for the input $x(t) = 50\cos 10t$.

OBJECTIVE 11–4 NETWORK FUNCTIONS AND CONVOLUTION (SECT. 11–6)

(a) Given the impulse response of a linear circuit, use the convolution integral to find the response to a specified input.

(b) Use the convolution integral to derive properties of linear circuits.

See Examples 11–15, 11–16, 11–17, 11–18 and Exercises 11–17, 11–18, 11–19, 11–20

11–38 The impulse response of a linear circuit is $h(t) = [u(t) - u(t-2)]$. Use the convolution integral to find the response to the input $x(t) = u(t-2)$.

11–39 The impulse response of a linear circuit is $h(t) = [u(t) - u(t-1)]$. Use the convolution integral to find the response to the input $x(t) = u(t) - u(t-2)$.

11–40 The impulse response of a linear circuit is $h(t) = 2t[u(t) - u(t-1)]$. Use the convolution integral to find the response to the input $x(t) = u(t-1)$.

11–41 The impulse response of a linear circuit is $h(t) = e^{-t}u(t)$. Use the convolution integral to find the response to the input $x(t) = u(t)$.

11–42 The impulse response of a linear circuit is $h(t) = t[u(t) - u(t-1)]$. Use the convolution integral to find the response to the input $x(t) = e^{-t}u(t)$.

11–43 The impulse response of a linear circuit is $h(t) = 2e^{-t}u(t)$. Use the convolution integral to find the response to the input $x(t) = tu(t)$.

11–44 Show that $f(t) * \delta(t) = f(t)$. That is, show that convolving any waveform $f(t)$ with an impulse leaves the waveform unchanged.

11–45 Show that if $h(t) = u(t)$, then output $y(t)$ for any input $x(t)$ is

$$y(t) = \int_0^t x(\tau)\, d\tau$$

That is, a circuit whose impulse response is a step function operates as an integrator.

11–46 Use the convolution integral to show that if the input to a linear circuit is $x(t) = u(t)$ then

$$y(t) = g(t) = \int_0^t h(\tau) d\tau$$

That is, show that the step response is the integral of the impulse response.

11–47 If the input to a linear circuit is $x(t) = tu(t)$, then the output $y(t)$ is called the ramp response. Use the convolution integral to show that

$$\frac{dy(t)}{dt} = \int_0^t h(\tau) d\tau = g(t)$$

That is, show that the derivative of the ramp response is the step response.

11–48 The impulse response of a linear circuit is $h(t) = u(t)$. Use MATLAB to compute the convolution integral and find the response to the input $x(t) = t[u(t) - u(t-1)]$.

11–49 The impulse response transform of a linear circuit is $H(s) = 25/(s+5)$ and $x(t) = tu(t)$. Use s-domain convolution to find the zero-state response $y(t)$.

11–50 The impulse responses of two linear circuits are $h_1(t) = e^{-2t}u(t)$ and $h_2(t) = 4e^{-4t}u(t)$. What is the impulse response of a cascade connection of these two circuits?

OBJECTIVE 11–5 NETWORK FUNCTION DESIGN (SECT. 11–7)

(a) Visualize design specifications using MATLAB.
(b) Design alternative circuits that realize a given network function and meet other stated constraints.
(c) Validate alternative designs using OrCAD simulation software.
(d) Evaluate alternative designs using stated criteria and select the best design.

See Examples 11–19, 11–20, 11–21, 11–22, 11–23, 11–24, 11–25 and Exercises 11–21, 11–22, 11–23, 11–24, 11–25.

11–51 D Design a circuit to realize the transfer function below using only resistors, capacitors, and OP AMPs. Scale the circuit so that all resistors are greater than 10 kΩ and all capacitors are less than 1 μF.

$$T_V(s) = \pm\frac{2 \times 10^5}{(s+100)(s+10{,}000)}$$

11–52 D Design a circuit to realize the transfer function below using only resistors, capacitors, and no more than one OP AMP. Scale the circuit so that all capacitors are exactly 10 nF.

$$T_V(s) = \pm\frac{100\,(s+1000)}{(s+100)(s+10{,}000)}$$

11–53 D Design a circuit to realize the transfer function below using only resistors, capacitors, and no more than one OP AMP. Scale the circuit so that the final design uses only 20-kΩ resistors.

$$T_V(s) = \pm\frac{10{,}000s}{(s+1000)(s+5000)}$$

11–54 D Design a *passive* circuit to realize the transfer function below using only resistors, capacitors, and inductors. Scale the circuit so that all inductors are 50 mH or less.

$$T_V(s) = \frac{s^2}{(s+2000)^2}$$

11–55 D A circuit is needed to realize the transfer function

$$T_V(s) = \pm\frac{(s+100)\,(s+500)}{(s+200)\,(s+1000)}$$

(a) Design the circuit using two OP AMPs.
(b) Design the circuit using only one OP AMP.
(c) Design the circuit using no OP AMPs.
In all cases, scale the circuit so that all parts use practical values.

11–56 D Design a circuit to realize the transfer function below using only resistors, capacitors, and OP AMPs. Scale the circuit so that all capacitors are exactly 100 nF.

$$T_V(s) = -\frac{400\,(s+100)}{s(s+10{,}000)}$$

11–57 ⟨D⟩ A circuit is needed to realize the impulse response transform shown below. Scale the circuit so that all parts use practical values.

$$H(s) = \pm \frac{200s + 10^6}{s^2 + 200s + 10^6}$$

11–58 ⟨E⟩ It is claimed that both circuits in Figure P11–58 realize the transfer function

$$T_V(s) = K\left(\frac{s + 2000}{s + 1000}\right)$$

(a) Verify that both circuits realize the specified $T_V(s)$.
(b) Which circuit would you choose if the output must drive a 1-kΩ load?
(c) Which circuit would you choose if the input comes from a 50-Ω source?
(d) It is further claimed that connecting the two circuits in cascade produces an overall transfer function of $[T_V(s)]^2$ no matter which circuit is the first stage and which is the second stage.
Do you agree or disagree? Explain.

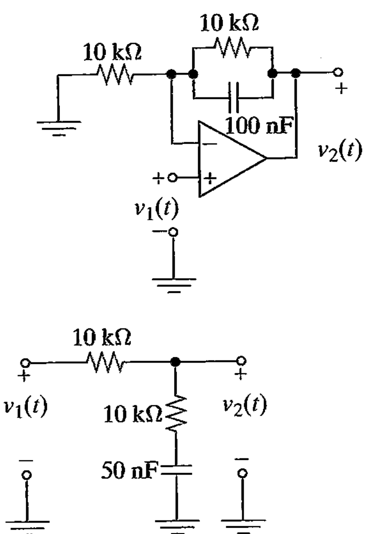


FIGURE P11–58

11–59 ⟨E⟩ It is claimed that both circuits in Figure P11–59 realize the transfer function

$$T_V(s) = \frac{\pm 1000s}{(s + 1000)(s + 4000)}$$

(a) Verify that both circuits realize the specified $T_V(s)$.
(b) Which circuit would you choose if the output must drive a 1-kΩ load?
(c) Which circuit would you choose if the input comes from a 50-Ω source?
(d) It is further claimed that connecting the two circuits in cascade produces an overall transfer function of $[T_V(s)]^2$ no matter which circuit is the first stage and which is the second stage.
Do you agree or disagree? Explain.

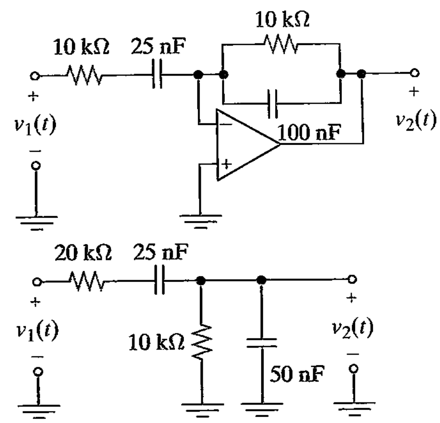


FIGURE P11–59

11–60 ⟨D⟩ Design a circuit that produces the following step response:

$$g(t) = 10\left[1 - e^{-50t} - 50te^{-50t}\right]u(t)$$

11–61 ⟨D⟩ A circuit is needed that will take an input of $v_1(t) = 2e^{-100t}u(t)$ V and produce an output of $v_2(t) = 5e^{-1000t}u(t)$ V. Design such a circuit using practical part values. Validate your design by using OrCAD.

11–62 ⟨D⟩ A circuit is needed that will take an input of $v_1(t) = \left[1 - e^{-10{,}000t}\right]u(t)$ V and produce a constant −5 V output. Design such a circuit using practical part values. Validate your design using OrCAD.

INTEGRATING PROBLEMS

11–63 ⟨A⟩ ⟨D⟩ First-Order Circuit Impulse and Step Responses
Each row in the table shown in Figure P11–63 refers to a first-order circuit with an impulse response $h(t)$ and a step response $g(t)$. Fill in the missing entries in the table.

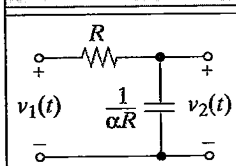
Circuit	$h(t)$	$g(t)$
		
	$\delta(t) - [\alpha e^{-\alpha t}]u(t)$	
		$\left(1 + \frac{e^{-\alpha t}}{2}\right)u(t)$

FIGURE P11–63

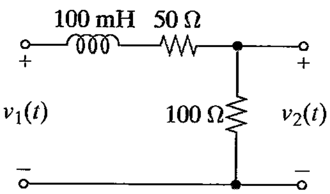


FIGURE P12–1

12–2 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12–2.

(a) Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.

(b) Sketch the straight-line approximation of the gain response.

(c) Calculate the gain at $\omega = 0.5\omega_C$, ω_C, and $2\omega_C$.

(d) Use OrCAD to plot the Bode magnitude gain response of the circuit. Validate your answers for part (c).

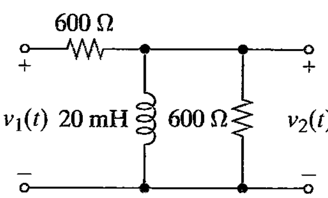


FIGURE P12–2

12–3 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12–3.

(a) Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.

(b) Sketch the straight-line approximation of the gain response.

(c) Calculate the gain at $\omega = 0.5\omega_C$, ω_C, and $2\omega_C$.

(d) Use OrCAD to plot the Bode magnitude gain response of the circuit. Validate your answers for part (c).

(e) What element values would you change to increase the passband gain to 10?

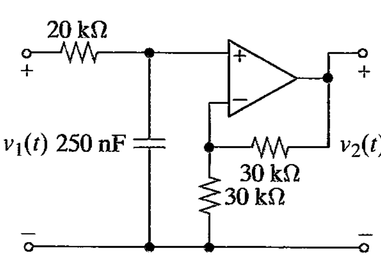


FIGURE P12–3

12–4 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12–4.

(a) Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.

(b) Sketch the straight-line approximation of the gain response.

(c) Calculate the gain at $\omega = 0.5\omega_C$, ω_C, and $2\omega_C$.

(d) Use OrCAD to plot the Bode magnitude gain response of the circuit. Validate your answers for part (c).

(e) What element values would you change to double the passband gain without changing the cutoff frequency?

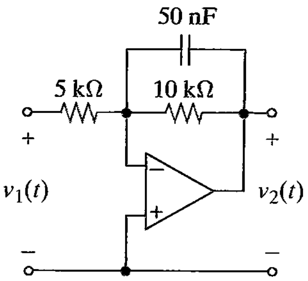


FIGURE P12–4

12–5 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12–5.

(a) Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.

(b) Use OrCAD to plot the Bode magnitude gain response of the circuit.

(c) What element value would you change to increase the cutoff frequency by one decade?

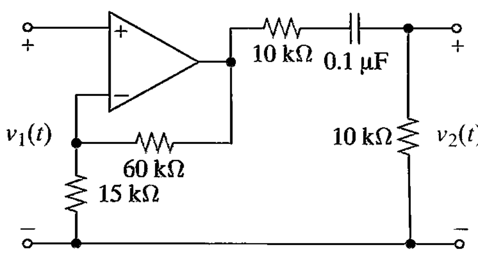


FIGURE P12–5

12–6 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12–6. What type of gain response does the circuit have? What is the passband gain? Select values of R and C so that the cutoff frequency is 10 kHz.

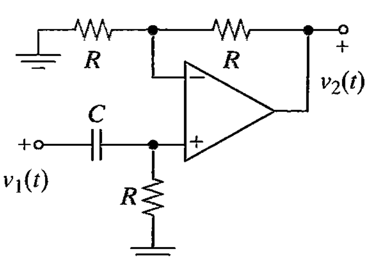


FIGURE P12–6

12–7 A first-order high-pass circuit has a passband gain of 20 dB and a cutoff frequency of 500 rad/s. Find the gain (in dB) at $\omega = 0, 200, 500, \text{and } 1000$ rad/s.

12–8 A first-order low-pass circuit has a passband gain of 5 dB and a cutoff frequency of 2 krad/s. Find the gain (in dB) at $\omega = 0, 200$ rad/s, 20 krad/s, and 200 krad/s.

12–9 **D** The transfer function of a first-order circuit is

$$T(s) = \frac{10{,}000}{s + 1000}$$

(a) Identify the type of gain response. Find the cutoff frequency and the passband gain.
(b) Use MATLAB to plot the magnitude of the Bode gain response.
(c) Design a circuit to realize the transfer function.
(d) Use OrCAD to validate your circuit design by comparing its frequency response with the MATLAB output.

12–10 **D** The transfer function of a first-order circuit is

$$T(s) = \frac{s}{s + 3000}$$

(a) Identify the type of gain response. Find the cutoff frequency and the passband gain.
(b) Use MATLAB to plot the magnitude of the Bode gain response.
(c) Design a circuit to realize the transfer function.
(d) Use OrCAD to validate your circuit design by comparing its frequency response with the MATLAB output.

12–11 **D** Design a passive first-order low-pass filter with a dc gain of 0 dB and a cutoff frequency of 5 kHz. Use OrCAD to validate your response.

12–12 **D** Design a first-order high-pass filter with a high-frequency gain of 50 and a cutoff frequency of 500 Hz. Use OrCAD to validate your response.

12–13 **D** Design a first-order low-pass filter with a dc gain of 2 and a gain of −3 dB at 1000 rad/s. Use OrCAD to validate your response.

OBJECTIVE 12–2 BANDPASS AND BANDSTOP RESPONSES (SECT. 12–4)

Given a cascade or parallel connection of two first-order circuits:
(a) Find and classify the frequency response.
(b) Plot the gain and phase responses using straight-line approximations and computer tools.
(c) Design circuits to produce a specified frequency response.
See Examples 12–8, 12–9 and Exercises 12–9, 12–10, 12–11

12–14 **D** The circuit in Figure P12–14 produces a bandpass response for a suitable choice of element values. Identify the elements that control the two cutoff frequencies. Select the element values so that the passband gain is 30 and the cutoff frequencies are 100 rad/s and 2500 rad/s. Use prac element values with $R \geq 10\,\text{k}\Omega$ and $C \leq 1\,\mu\text{F}$. Use OrC to validate your design.

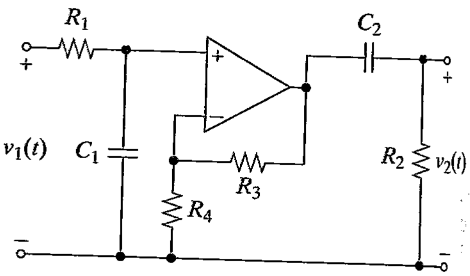


FIGURE P12–14

12–15 **D** Repeat Problem 12–14 for the circuit in Figure P12–1 Use OrCAD to validate your design.

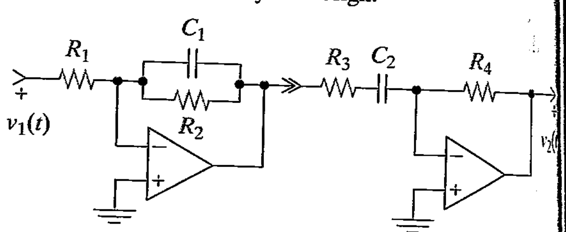


FIGURE P12–15

12–16 **E** Suppose the circuits you designed in Problem 12–14 and 12–15 had to feed a 500-Ω recorder. Which circu would you select and why?

12–17 **D** The circuit in Figure P12–17 produces a bandsto response for a suitable choice of element values.
(a) Find the circuit's transfer function.
(b) Identify the elements that control the two cutoff fre quencies. Select the element values so that the cutoff fre quencies are 100 krad/s and 1000 krad/s. Use practica element values with $R \geq 10\,\text{k}\Omega$, $L \leq 100\,\text{mH}$, and $C \leq 1\,\mu\text{F}$
(c) Use MATLAB to plot the Bode magnitude plot for the values you selected.
(d) What is the passband gain for your design?
(e) Simulate your circuit using OrCAD and compare the results with the MATLAB output.

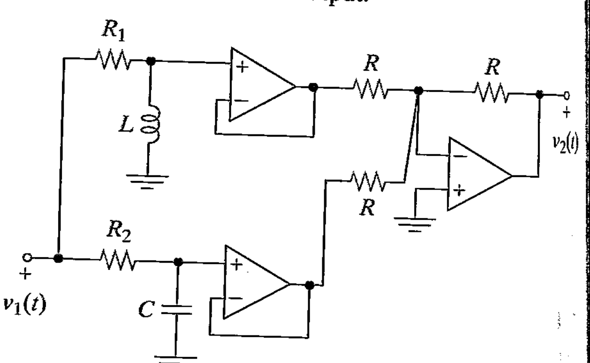
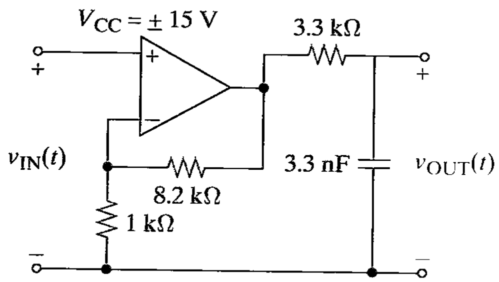


FIGURE P12–17

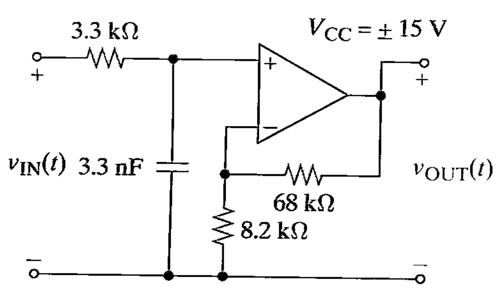
engineer, the project manager asks you evaluate the designs and recommend a vendor. Which vendor would you recommend and why?

First-Order Filter Company



Unit Cost: $29.00

Simply Filters, Ltd



Unit Cost: $20.99

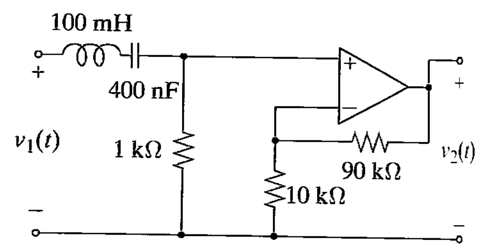
FIGURE P12–56

12–57 **E** Design Evaluation

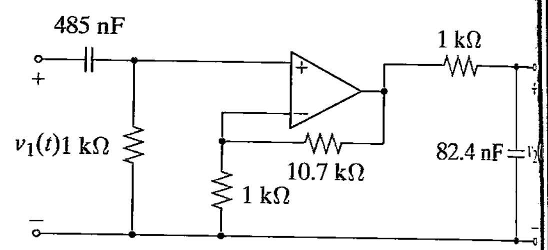
In a research laboratory, you need a bandpass filter to meet the following requirements:

Design requirements: Passband gain: 10 ± 5%, $B = 10$ krad/s $\pm 5\%$, $\omega_0 = 5$ krad/s $\pm 2\%$, $\omega_{CL} = 2$ krad/s $\pm 10\%$.

Evaluation criteria: Filter performance, parts count, use of standard parts, and cost. The two vendors have responded with the designs shown in Figure P12–57. As a research assistant, your supervisor asks you to evaluate the two designs and recommend a vendor. Which vendor would you recommend and why?



Vendor #1



Vendor #2

FIGURE P12–57